A Watershed Moment

The Clean Water Act and Birth Weight

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ABSTRACT

The Clean Water Act (CWA) significantly improved surface water quality, but at a cost exceeding the estimated benefits. We quantify the effect of the CWA on a direct measure of health. Using a difference-in-differences framework, we compare birth weight upstream and downstream from wastewater treatment facilities before and after CWA grant receipt. Pollution only decreased downstream from facilities required to upgrade their treatment technology, and we leverage this additional variation with a triple difference. Clean Water Act grants increased average birth weight by 8 grams.

I. Introduction

The Clean Water Act is a landmark, yet controversial, policy. The CWA caused significant reductions in pollution, but improvements in surface water quality stemming from the CWA have come at a high cost. Projects funded through grants to wastewater treatment facilities between 1960 and 2005 have cost about $870 billion over their lifetimes (in 2017 dollars) (Keiser and Shapiro 2019b). In total, U.S. government and industry have spent more than $1.9 trillion to abate surface water pollution (Keiser, Kling, and Shapiro 2019). Existing analyses of the Clean Water Act estimate benefits that are lower than the act's costs (for example, Lyon

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and Farrow 1995; Freeman 2010; Keiser, Kling, and Shapiro 2019), but these analyses have not generally included improvements in health caused by the Clean Water Act because there has not been a systematic ex post measurement of the health benefits of the CWA. To our knowledge, this paper is the first to estimate the effect of CWA grants on birth weight.

Estimating the health benefits of CWA grants may matter for several reasons. Historically, policies targeting improvements in child health generate high returns to public funds (Hendren and Sprung-Keyser 2020), and previous economics literature shows that even small differences in child and infant health can lead to large impacts on later life outcomes (Behrman and Rosenzweig 2004; Royer 2009; Black, Devereux, and Salvanes 2007; Figlio et al. 2014; Isen, Rossin-Slater, and Walker 2017; Black et al. 2019). Health benefits often account for a large portion of the total benefits of environmental regulation, with health effects accounting for more than 95 percent of all benefits of air pollution regulation (Keiser, Kling, and Shapiro 2019).

Existing economics research estimates the benefits of improved surface water by measuring the effect of CWA grants on nearby housing prices, but this hedonic analysis may not fully capture health effects of CWA grants. Comparing waters up- and downstream from wastewater treatment facilities, Keiser and Shapiro (2019a) find that CWA grants caused reductions in downstream pollution. These improvements in water quality were capitalized into housing prices, but increases in home values were substantially smaller than the CWA's costs. By quantifying how residents value water quality, Keiser and Shapiro (2019a) improve upon previous cost–benefit calculations, but as noted in Keiser, Kling, and Shapiro (2019), hedonic analysis assumes housing values reflect the implicit value that households place on the quality of nearby surface water. If households are uninformed about nearby surface water quality or do not understand the benefits of reduced surface water pollution, housing values will not reflect the health benefits of the program. In this historic context, it is unlikely that households fully understood the range and extent of any negative health effects of surface water contamination, especially the negative impacts on developing fetuses in utero. By directly estimating the health effects of the CWA, our results complement those of Keiser and Shapiro (2019a) by quantifying one of the largest benefits of the CWA that hedonic analysis is least likely to capture.

Using a difference-in-differences design, we compare birth weight upstream and downstream from wastewater treatment facilities before and after the facility receives a CWA grant. Comparing up- and downstream births addresses the endogenous distribution of grants, as well as any economic shocks caused by grant receipt, but estimates may still be biased if individuals sort into downstream areas or if these areas experience differential trends relative to upstream areas after grant receipt. To address this concern, we show that CWA grants only caused improvements in surface water quality downstream from facilities that were required to upgrade their treatment technology to comply with new treatment technology standards imposed by the CWA. This finding motivates a triple difference design that uses counties up- and downstream from facilities where these treatment technology requirements were not binding as an additional control group. By using already compliant facilities that receive grants as an additional control group, we can account for differential sorting into downstream areas after grant receipt, so the health benefits we capture with this design are likely caused by improvements in water quality.

Across specifications, we consistently find that CWA grants had a statistically significant impact on downstream birth weight. Our results show that reductions in surface water pollution from the CWA are associated with an 8-gram increase in average birth weight.

Our results contribute to a literature documenting the importance of effective sewerage and clean water in protecting health historically in the United States (Troesken 2001, 2002; Cutler and Miller 2005; Ferrie and Troesken 2008; Beach et al. 2016; Anderson, Charles, and Rees 2022), as well as a literature on the complementarity of sewerage and clean water interventions. Examining water policy in early 20th-century Massachusetts, Alsan and Goldin (2019) show that mortality declines were driven by a combination of clean water initiatives and effective sewerage. Watson (2006) shows that federal sanitation policies explain much of the convergence in Native American and white infant mortality rates in the United States since 1970. By improving sewerage systems and reducing pollution of surface water throughout the United States at a time when most publicly provided drinking water had some treatment, the CWA provides a new context to examine the effect of improved water quality on health in the late 20th century, as well as the complementarity between sewerage infrastructure and clean water in protecting health nationwide.

II. Background

The Clean Water Act aimed to slow the flow of contaminants from point sources, such as municipal waste treatment facilities and industrial pollution sources, into rivers and lakes.[1] This work focuses on CWA grants distributed to municipal wastewater treatment facilities. Wastewater from homes, businesses, and industries, as well as surface runoff, is typically collected through a system of sewers and delivered to a wastewater treatment facility for treatment and discharge into local waterways (USEPA 2004). To reduce surface water contamination, the CWA addressed pollution from municipal waste treatment plants with two complementary policies: grants to wastewater treatment facilities and regulation of wastewater treatment technology.[2]

A. Grants

From 1972 to 1988, the U.S. Environmental Protection Agency (EPA) distributed an estimated $153 billion (in 2014 dollars) in grants to municipal governments for capital upgrades to wastewater treatment facilities. The EPA allocated CWA grant money to states according to a formula based on total population, forecast population, and wastewater treatment needs (Rubin 1985). States then distributed grants to municipalities

1. Although much of the contamination of U.S. waterways comes from sources that cannot be traced back to a specific facility, such as agricultural runoff, the Clean Water Act did not directly regulate these “nonpoint” pollution sources. The CWA did not directly regulate drinking water supplies either. The Safe Drinking Water Act sets minimum standards for drinking water quality for all public water systems in the United States.

2. In addition to regulating municipal waste treatment facilities, the CWA required all industrial polluters to obtain a permit from the National Pollutant Discharge Elimination System (NPDES) before discharging wastewater. Regulation through the NPDES led to reductions in both profits (Rassier and Earnhart 2010) and the number of environmental employees (Raff and Earnhart 2019) at newly regulated polluters.

according to priority lists based on the severity of nearby surface water pollution, the size of the population affected, the need for conservation of the affected waterway, and that waterway's specific category of need (USEPA 1980).

Since state governments wrote their own priority lists, grant placement may be correlated with trends in infant health. Moreover, grants could cause increases in birth weight that are unrelated to changes in pollution by improving economic conditions with an influx of federal dollars. Instead of treating grant timing and location as exogenous, we compare the difference in birth outcomes in areas up- and downstream from a given wastewater treatment facility before and after grant receipt between facilities that were required to make treatment technology upgrades and all other facilities. To the extent that other policies were changing during this time period, and that grants improved local economic conditions, these changes were likely to affect upstream and downstream areas similarly.

B. Regulation

In 1972, about a quarter of all U.S. municipal wastewater treatment facilities reported using relatively inexpensive, but less effective, primary treatment (USEPA 2000). This process, depicted in Online Appendix Figure A1a, forces wastewater through a series of screens. While primary treatment removes large detritus and heavy biosolids, it still discharges all but the heaviest organic material into waterways (USEPA 1998).

The Clean Water Act required all municipal treatment plants to upgrade to secondary treatment. Plants use secondary treatment technology, shown in Online Appendix Figure A1b, in addition to primary treatment. After screens filter out large debris, wastewater sits in an aeration tank where bacteria in the water consume organic material, which ultimately reduces biochemical oxygen demand (BOD).[3]

Municipal waste is almost entirely organic (Hines 1966) and often contains both pathogenic and nonpathogenic microorganisms harmful to human health. Pathogens harmful to human health include enteric bacteria, viruses, protozoa, parasitic worms, and their eggs. These microorganisms from human sewage can cause a range of gastrointestinal illnesses and infections (Reynolds, Mena, and Gerba 2008; Chahal et al. 2016). Even today, estimates suggest that up to 19.5 million cases of waterborne illnesses are associated with contaminated drinking water each year in the United States (Reynolds, Mena, and Gerba 2008; Colford et al. 2006). Online Appendix Table A1 reports pathogens capable of causing waterborne or water-based disease. The potential health effects from exposure to waterborne or water-based diseases include gastroenteric infections and diseases, with symptoms such as nausea, vomiting, diarrhea, and stomach cramps. Exposure to waterborne contaminants can result from recreational use of affected surface water or from ingestion of contaminated public or private drinking water sources. Pregnant women are considered a high-risk population, and infant health may be impacted either directly through impacts on fetal development or indirectly through maternal illness that may result in reduced nutritional intake, for example.

3. Additionally, some states required facilities to meet more stringent treatment technology requirements than the CWA's mandate, such as tertiary treatment, which is aimed at removing ammonium, nitrate, and phosphate (USEPA 2000).

Through biological oxidation, secondary treatment can remove more than 90 percent of harmful pathogenic bacteria and viruses from sewage (Abdel-Raouf, Al-Homaidan, and Ibraheem 2012).[4] While it is not practical to monitor pathogens directly, regulators and researchers often use indicator organisms, such as total or fecal coliforms, to monitor water quality. Keiser and Shapiro (2019a) show that grants to wastewater treatment facilities improved key indicators of water quality, including dissolved oxygen deficit, BOD, and fecal coliforms. As dissolved oxygen deficit is the most consistently and widely monitored measure of water quality in our sample, we focus on this measure.

The potential benefits of upgrading a facility's treatment technology were well understood, but waste treatment capital upgrades were expensive. Upgrading to secondary treatment technology could increase a facility's operating costs by up to 60 percent and require capital investments of as much as 30 percent of the initial cost of the facility (National Environmental Research Center 1972). Because of these costs, many treatment plants had not yet installed secondary treatment technology at the time of CWA federal mandate. Only about 47 percent of plants were in already in compliance because they had already installed secondary or more stringent treatment technology, typically due to historical pressure from downstream communities to reduce the flow of harmful pollutants and stronger state regulation of surface water pollution (Stoddard et al. 2003).[5] The remaining 53 percent of plants, as measured in the 1972 Clean Watershed Needs Survey (CWNS), were not yet in compliance with the relevant treatment technology mandates.[6] Treatment plants that were not already in compliance with both state and federal capital mandates in 1972, which we refer to as "noncompliant" facilities, had a strong incentive to use CWA grants to offset the costs of upgrading their treatment technology.[7]

Many facilities that were already in compliance with both state and federal mandates still received CWA grants. While these facilities could make capital improvements, such as increasing capacity, they had less incentive to do so. Since the CWA did not mandate these upgrades, there was no binding constraint requiring these facilities to spend grant money on sewerage capital upgrades, and the municipalities that operated them faced pressure to use grant money to offset the operating costs of their water and sewerage utilities in an attempt to lower costs for consumers and become more competitive (Daigger 1998).[8]

Since noncompliant facilities had a clear channel through which to improve surface water quality and were more likely to spend CWA grant money on capital upgrades, we expect the reductions in downstream pollution associated with CWA grants to be

4. Suspended (for example, activated sludge) growth reactors remove about 90 percent of viruses, but removal can be more varied in film reactors, which provide less absorption (Abdel-Raouf, Al-Homaidan, and Ibraheem 2012).

5. We show that outcomes in these compliant and noncompliant cities trend similarly prior to grant receipt in our event study models.

6. See Online Appendix Section C.3 for more discussion.

7. Permits distributed to polluters through the NPDES required municipal treatment plants to satisfy the treatment technology mandate, and violators could be fined up to $25,000 per day (Copeland 2016).

8. Flynn and Smith (2022) show that CWA grants to noncompliant municipalities led to a dollar for dollar increase in sewerage capital spending, while grants to facilities already in compliance with state and federal capital mandates crowded out funds that municipalities were already spending on sewerage capital rather than causing an increase in sewerage capital spending.

largest for noncompliant facilities. This motivates a triple difference design that uses areas nearby facilities that were not indicated as pre-CWA noncompliant in the 1972 CWNS as an additional control group.

III. Data

A. CWA Grants and Municipal Wastewater Treatment Plants

We obtain data on all 33,429 grants that the EPA distributed to 14,285 wastewater treatment plants from the EPA's Grant Information Control System.[9] Most facilities received multiple grants, so we define a facility as "treated" after it receives its first CWA grant and show the distribution of facility-level treatment timing in Online Appendix Figure A2.

Using a unique facility code, we merge these grant data with the Clean Watershed Needs Survey, an assessment of the capital investment needed to meet the water quality goals of the CWA. These linked data include facility location, grant timing, and state and federal treatment technology compliance status as of 1972 (Marcus and Flynn 2023).[10]

B. Spatial Data on Waterways

We define treatment in terms of the flow direction of waterways. We determine if a waterway is up- or downstream from a facility with the National Hydrography Data Set, an electronic atlas that maps the location and flow direction of all U.S. waterways. We follow both the EPA and other researchers studying the Clean Water Act by focusing on areas 25 miles up- and downstream from treatment facilities (Keiser and Shapiro 2019a; USEPA 2001). Online Appendix Table B1 shows that our results are robust to concentrating on areas five or ten miles downstream from treatment facilities.

We define a county as downstream if it contains any waterway that is within 25 miles downstream of a treated facility. Therefore, we categorize counties with both up- and downstream waterways as downstream. Figure 1 provides maps showing two examples of how counties are classified based on upstream and downstream waterways. Waterways within 25 miles downstream of a treated facility are dark gray (blue online) and waterways 25 miles upstream are light gray (yellow online). Figure 1A shows an example from a wastewater facility in Indiana, where Shelby and Johnson counties would be classified as downstream because they contain a waterway within 25 miles downstream of a treated facility. Hancock county, on the other hand, would be classified as an upstream county because it contains only

9. The 33,429 grants in our sample exclude grants that do not include a specific facility code, as it is unclear to what extent these grants were precisely for wastewater treatment plants. Online Appendix Section C.3 provides further discussion.

10. There are 1,930 facilities in our analysis sample that are missing data on pre-CWA treatment technology. We assume that these facilities were already in compliance with state and federal treatment technology requirements. Throughout the paper, we refer to the set of "compliant" facilities, which includes all facilities that were not explicitly "noncompliant" in the 1972 CWNS. Our results are similar when we exclude facilities with missing information on treatment technology.

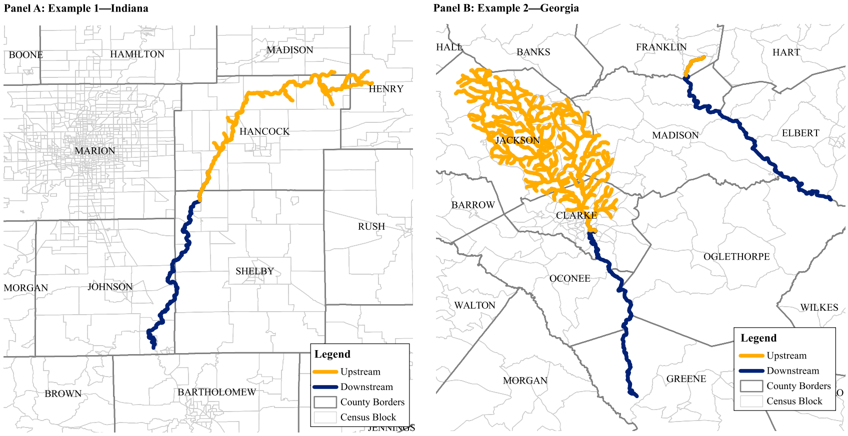

Figure 1

Defining Upstream and Downstream Counties

Notes: The figures show waterways within 25 miles downstream of a treated facility in dark gray (blue online) and 25 miles upstream in light gray (yellow online). County boundaries are in dark gray, and census block boundaries are in light gray.

upstream waterways. Figure 1B shows two wastewater treatment facilities in Georgia. One facility is located in Clarke County. For this facility, downstream counties include Clarke, Oconee, Oglethorpe, and Greene. In this case, Jackson is the only upstream county for this facility. Although Madison County contains upstream waterways for the facility located in Clarke County, it cannot be an upstream county because it contains downstream waterways for the other wastewater treatment facility and therefore is classified as a downstream county.

C. Water Pollution

Data on dissolved oxygen deficit come from STORET legacy, which includes readings from pollution monitoring stations across the United States.[11] We include readings from pollution monitors on rivers and lakes located 25 miles up- or downstream from any facility in the CWNS data. We also follow the data cleaning steps laid out in the appendix of Keiser and Shapiro (2019a).

D. Infant Health

We measure infant health with birth certificate data from the National Center for Health Statistics (NCHS) from 1968–1988 (National Center for Health Statistics

11. Dissolved oxygen deficit is a continuous measure defined as 100 minus dissolved oxygen saturation (dissolved oxygen level divided by water's maximum oxygen level). It is one of the most common measures of omnibus water pollution in research, and it responds to a wide variety of pollutants (Keiser and Shapiro 2019a).

1968–1988a). These data contain information on birth weight, birth order, mother's age and race, and county of residence for each birth.[12] Online Appendix Table A2 presents summary statistics for individual-level births in 1970, two years before the first CWA grants were distributed, from up- and downstream counties.

While ideal data would contain exact addresses, these data are unavailable for most states, and even when collected, addresses are typically not available until the after the adoption of the 1989 U.S. Standard Birth Certificate revision and the use of electronic birth certificates, which is after our study period.

We collapse birth data to county means, calculating the average birth weight, the probability of low birth weight, the percent of nonwhite births, average mother's age, and the probability of being a mother's first, second, third, or fourth or higher birth in each county–year.[13] Although more recent birth records data contain far more variables of interest, such as gestation, maternal education, and maternal risk factors, these variables are either unavailable or not reliably and consistently recorded in data from 1968–1988.

E. Population Density

We expect the health effects of improved surface water quality to be concentrated near treated waterways. County-level exposure depends on the distribution of the population within a county relative to the location of treated waterways. We use 1990 census block population density data from the U.S. Census Bureau to measure the percent of a county's population living within a mile of a treated waterway.[14] Assuming a uniform population distribution within census blocks, this provides a proxy for the probability that mothers within the county are exposed to treated waterways. Online Appendix Figure A3 shows the distribution of this treatment measure. We discuss how this measure captures different exposure pathways in Section IV.C.

IV. First Stage: Water Pollution

A. Methods

Before comparing birth outcomes up- and downstream from wastewater treatment facilities, we examine the first-stage relationship between grant receipt and downstream water quality with Equation 1.

12. Data before 1972 constitute a 50 percent random sample of all births in the United States. After 1972, some states report data on all births. Six states had full sample data in 1972, and all states and the District of Columbia had full sample data by 1985. Online Appendix C.1 provides additional information and shows our main results are not driven by sampling changes.

13. We also calculate county means of one year mortality using data from NCHS (National Center for Health Statistics 1968– 1988b). We find no significant effect of CWA grants on this outcome in Online Appendix Table A9; however, our estimates are imprecise.

14. We use data from 1990 because it is the first census for which population density data are available at the census block level. In Online Appendix B.3, we also show similar but attenuated results if we define treatment with a binary variable. We also show in Online Appendix Table A6 that the results are robust to dropping the largest quartile of counties by geographic area, as these large counties may have more measurement error.

$$(1) \quad Q_{pdy} = \gamma g_{py} * d_d + \beta W_{pdy} + \alpha_{py} + \alpha_{pd} + \varepsilon_{pdy}$$

Q_{pdy} is a measure of dissolved oxygen deficit, and g_{py} equals one after a facility receives its first CWA grant. There are two observations for each treatment plant p for each year y, which describe average dissolved oxygen deficit upstream ($d_d = 0$) and downstream ($d_d = 1$) from that plant. Since dissolved oxygen deficit varies inversely with temperature, W_{pdy} measures water temperature.

We include plant-by-downstream and plant-by-year fixed effects, α_{pd} and α_{py}, respectively. Plant-by-downstream fixed effects allow waters up- and downstream from a given wastewater treatment plant to have different mean levels of dissolved oxygen deficit, which controls for pollution sources located up- or downstream from a plant that are constant over time. Plant-by-year fixed effects ensure that we are only comparing waters up- and downstream from the same facility, which controls for any yearly shocks that affect waters both up- and downstream from a facility. All standard errors in our pollution estimates are clustered at the facility level.

We estimate Equation 1 for the full sample and subsamples of compliant and noncompliant facilities, as well as a fully interacted triple difference specification. These estimates give us a sense of how grants and regulations worked together by testing whether pollution evolved differently in waters downstream from noncompliant facilities and compliant facilities after grant receipt.

B. Pollution Results

Table 1 estimates the effect of CWA grant receipt on downstream water quality. Columns 1–3 present estimates of Equation 1 on the full sample, noncompliant facilities, and compliant facilities, respectively. Column 4 presents coefficients from a triple difference specification. As shown in Column 2, dissolved oxygen deficit only decreased significantly in water downstream from noncompliant facilities. Since dissolved oxygen deficit is defined as 100 minus dissolved oxygen saturation, this result show that, after grant receipt, dissolved oxygen saturation increased by 1.6 percentage point in waters downstream from noncompliant facilities relative to waters upstream from the same facility. The coefficient for waters downstream from compliant facilities in Column 3 is small and statistically insignificant. The reduction in dissolved oxygen deficit downstream from noncompliant facilities is statistically larger than the change downstream from compliant facilities, as shown by the significant negative triple difference coefficient in Column 4.

Figure 2A presents results from the event study corresponding to the triple difference in Column 4. This figure shows that reductions in downstream pollution were significantly larger in waters downstream from noncompliant facilities relative to waters downstream from compliant facilities. In addition, there does not appear to be any trend in pollution prior to grant receipt, which might have arisen from compliant facilities' early adoption of more advanced treatment technology. In the analysis of the impact of CWA grants on birth weight that follows, we leverage this comparison between noncompliant and compliant facilities in a triple difference specification.

Table 1
Effects on Surface Water Pollution

	Full Sample (1)	Noncompliant (2)	Compliant (3)	DDD (4)
Grant × downstream	−0.974***	−1.566**	−0.371	−0.371
	[−1.364, −0.584]	[−2.125, −1.008]	[−0.911, 0.170]	[−0.911, 0.170]
Grant × downstream × noncompliant				−1.196***
				[−1.973, −0.419]
Weather controls	X	X	X	X
Facility by downstream fixed effects	X	X	X	X
Facility by year fixed effects	X	X	X	X
N	114,148	46,968	67,180	114,148

Source: (USEPA 1968–1988).
Notes: This table describes the effects of Clean Water Act grants on downstream pollution. Columns 1, 2, and 3 estimate Equation 1 for areas up- and downstream from all facilities in our sample, noncompliant facilities, and all other facilities, respectively. Column 4 presents estimates from the associated triple difference (DDD): $Q_{pdy} = \gamma^{DD} g_y * d_d + \gamma^{DDD} g_y * d_d * t_p + \beta W_{pdy} + \varphi W_{pdy} * t_p + \alpha_{py} + \alpha_{pd} + \varepsilon_{pdy}$, where t_p is a dummy variable equal to one for observations from noncompliant facilities. Q_{pdy} is dissolved oxygen deficit, g_y is a dummy variable equal to one after a facility receives a CWA grant, and d_d is a dummy equal to one for observations downstream from a facility. All regressions include controls for water temperature, as well as facility-by-downstream and facility-by-year fixed effects, α_{pd} and α_{py}. Ninety-five percent confidence intervals in brackets. $*p < 0.10$, $**p < 0.05$, $***p < 0.01$.

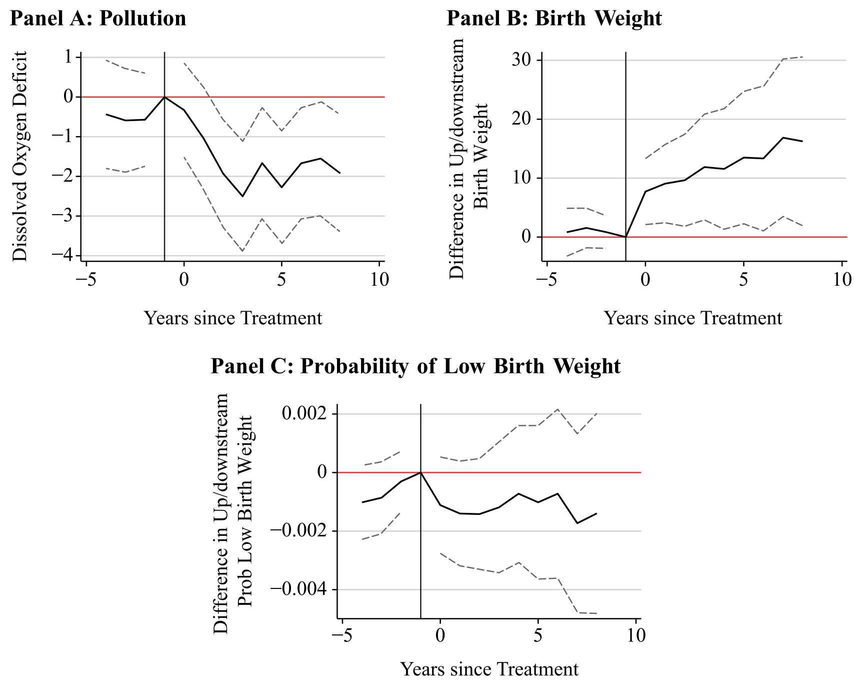

Figure 2

Pollution and Infant Health Event Studies

Source: USEPA (1968– 1988) and National Center for Health Statistics (1968–1988a).

Notes: Panel A plots the θ_t and η_t from estimating

$$Q_{pdy} = \sum_{t=-5}^{-2} \theta_t 1\{y - y_p^* = t\} * d_d * t_p + \sum_{t=0}^{9} \eta_t 1\{y - y_p^* = t\} * d_d * t_p + \sum_{t=-5}^{-2} \pi_t 1\{y - y_p^* = t\} * d_d$$
$$+ \sum_{t=0}^{9} \gamma_t 1\{y - y_p^* = t\} * d_d + \beta W_{pdy} + \varphi W_{pdy} * t_p + \alpha_{py} + \alpha_{pd} + \varepsilon_{pdy}.$$

Q_{pdy} measures dissolved oxygen deficit, d_d is a dummy equal to one for observations downstream from a facility, and t_p is an indicator that equals one for noncompliant facilities. The model includes facility-by-downstream fixed effects and facility-by-year fixed effects, α_{pd} and α_{py}, as well as controls for temperature. Panels B and C plot the θ_t and η_t from estimating

$$\Delta Y_{py} = \sum_{t=-5}^{-2} \theta_t 1\{y - y_p^* = t\} * d_d * t_p + \sum_{t=0}^{9} \eta_t 1\{y - y_p^* = t\} * pct_{py} * t_p + \sum_{t=-4}^{-2} \pi_t 1\{y - y_p^* = t\}$$
$$+ \sum_{t=0}^{9} \gamma_t 1\{y - y_p^* = t\} * pct_{py} + \beta X_{py} + \varphi X_{py} * t_p + \alpha_y * t_p + \alpha_p + \alpha_y + \varepsilon_{py}.$$

pct_{py} is a continuous variable that takes values from zero to one, and indicates the percent of downstream counties' populations living within a mile of a treated waterway in year y. The model includes facility and year fixed effects, α_p and α_y, respectively, as well as demographic controls. t_p is an indicator that equals one for noncompliant facilities. The dependent variable is the difference in birth weight between up- and downstream counties in year y in Panel B and the difference in the probability of being born weighing less than 2,500 grams between up- and downstream counties in year y in Panel C.

V. Infant Health

A. Methods

We begin our reduced-form analysis of the impact of CWA grants on birth weight by comparing birth outcomes in counties downstream from treated facilities to all other areas with the following difference-in-differences specification

$$(2) \quad Y_{cy} = \gamma pct_{cy} + \beta X_{cy} + \alpha_c + \alpha_y + \varepsilon_{cy}$$

Y_{cy} is an average birth outcome in county c in year y, and pct_{cy} is the percent of county c's population living within a mile of a treated waterway in year y.[15] Controls in X_{cy} include the percent of births that were a mother's first, second, third, or fourth, and county averages of mother's age and race. α_c and α_y are county and year fixed effects. Observations are at the county–year level, and standard errors are clustered at the county level. Since we collapse birth weight data to county means, we weight our results by the total number of births that occurred in a county–year.

The presence of local area trends specific to a facility's location could mean that an upstream county is only a good counterfactual for a county located downstream from the same facility. We address this concern in our next specification by collapsing our data to the facility rather than the county level. Our outcome variable is now ΔY_{py}, which is equal to the mean birth weight in all counties downstream from a facility minus the mean birth weight in all counties upstream from the same facility in each year. We then estimate the following specification

$$(3) \quad \Delta Y_{py} = \gamma pct_{py} + \beta X_{py} + \alpha_p + \alpha_y + \varepsilon_{py}$$

where p indexes facilities, and pct_{py} measures the percent of downstream counties' populations living within a mile of a treated waterway. We include facility and year fixed effects, α_p and α_y, respectively.[16] Standard errors are clustered at the facility level.

This specification requires us to assume that, in the absence of grant receipt, birth outcomes would have evolved similarly in areas up- and downstream from the same facility after grant receipt. This assumption would be violated if, for example, downstream areas were experiencing differential sorting patterns or greater economic growth relative to upstream areas, even in the absence of CWA grants. For example, Keiser and Shapiro (2019a) show that downstream housing prices increase after grant receipt, which may cause healthier mothers to sort into downstream communities.

15. By using the percent of a county's population living within a mile of a treated waterway as a proxy for the probability that mothers within the county are exposed to treated waterways, we employ a continuous measure of treatment. To identify the average treatment effect, we must assume that, for all doses, the average change in outcomes over time across all units if they had been assigned a particular dose is the same as the average changes in outcomes over time for all units that did experience that dose (Callaway, Goodman-Bacon, and Sant'Anna 2021). This assumption could be violated if treatment effects vary over time or if there is selection into a given dose. We address the dynamic treatment effects in Online Appendixes B.2 and B.3. As the dose in this setting is simply a measure of what portion of births in a county are likely to be treated, we think the strong parallel trends assumption is reasonable.

16. Controls in facility-level specifications are averages from all births in up- and downstream counties. Our results are robust to controlling for the difference between average demographic characteristics in up- and downstream counties instead.

To address concerns regarding differential trends in birth weight in downstream relative to upstream areas caused by differences in economic growth or sorting of households into downstream areas, we employ a triple difference design. We estimate the following equation

$$(4)\quad \Delta Y_{py} = \gamma^{DD} pct_{py} + \gamma^{DDD} pct_{py} * t_p + \beta X_{py} + \varphi X_{py} * t_p + \alpha_y * t_p + \alpha_p + \alpha_y + \varepsilon_{py}$$

where t_p is an indicator that equals one for noncompliant facilities. In this specification, the first difference comes from where and when CWA grants were distributed, the second comes from if a birth occurred up- or downstream from a wastewater treatment facility, and the third comes from the facility's compliance with the treatment technology mandate.

Even if individuals sort into downstream communities, so long as the sorting pattern induced by grant receipt is similar for both compliant and noncompliant facilities, using compliance as a third difference will capture unobserved changes to up- and downstream counties occurring contemporaneously with CWA grant receipt. We test this by exploring how maternal characteristics evolve after grant receipt in upstream and downstream areas across noncompliant and compliant facilities. Table 2 estimates Equation 2 on demographic characteristics that are correlated with birth weight, such as race, age, and birth order. Column 1 of Table 2 shows results for the subsample of noncompliant facilities, and Column 2 shows results for compliant facilities. Column 3 presents results from the associated triple difference. Columns 1 and 2 show that areas downstream from facilities that received CWA grants had smaller nonwhite populations, slightly older mothers, and fewer higher-order births, but changes in downstream demographic characteristics are very similar across noncompliant and compliant facilities. The triple difference coefficients presented in Column 3 are small and statistically insignificant for all observed demographic outcomes, indicating that there was no observable differential sorting into downstream areas across noncompliant and compliant facilities after grant receipt. While we also control for these observable demographic characteristics in all regressions, these results provide some evidence that the identification assumption for the triple difference specification is likely to hold for unobservable characteristics as well.

B. Infant Health Results

Panel A of Table 3 shows effects on birth weight that are robust across a variety of specifications. Column 1 compares births in counties downstream from grant facilities to those in any other county by estimating Equation 2 using a sample of births from every county in the contiguous United States. Column 2 adds demographic controls to this specification.[17] Since births occurring in counties that are not near wastewater treatment facilities might not make a good control group, Column 3 excludes counties that are not up- or downstream from any wastewater treatment facility. This compares births in a downstream county to those in any upstream county. The results are similar to those from the full sample.

Counties upstream from the same facility are likely to make better counterfactuals for downstream counties than counties upstream from any facility. Column 4 estimates

17. Online Appendix Figure A4 shows the event study figures that correspond to the estimate in Column 2.

Table 2
Effects on Demographic Changes

	Noncompliant (1)	Compliant (2)	DDD (3)
Panel A: Percent Nonwhite			
Percent downstream	−0.0223***	−0.0176***	−0.0176***
	[−0.0281, −0.0165]	[−0.0229, −0.0123]	[−0.0229, −0.0123]
Percent downstream × noncompliant			−0.00471
			[−0.0126, 0.00313]
Mean	0.116	0.105	0.11
Panel B: Mother's Age			
Percent downstream	0.126***	0.0784**	0.0784**
	[0.0557, 0.197]	[0.0149, 0.142]	[0.0150, 0.142]
Percent downstream × noncompliant			0.0479
			[−0.0470, 0.143]
Mean	24.563	24.569	24.566
Panel C: Probability First or Second Birth			
Percent downstream	−0.00210	0.00109	0.00109
	[−0.00916, 0.00496]	[−0.00390, 0.00608]	[−0.00390, 0.00608]
Percent downstream × noncompliant			−0.00319
			[−0.0118, 0.00545]
Mean	0.653	0.645	0.648
Panel D: Probability Third or Higher Birth			
Percent downstream	−0.0105***	−0.00618***	−0.00618***
	[−0.0145, −0.00646]	[−0.00965, −0.00271]	[−0.00964, −0.00271]
Percent downstream × noncompliant			−0.00429
			[−0.00958, 0.00100]
Mean	0.338	0.347	0.343
Unit and year fixed effects	X	X	X
Collapsed to facility level	X	X	X
N	34,188	48,132	82,320

Source: National Center for Health Statistics (1968–1988a).
Notes: Columns 1 and 2 present results from estimating $\Delta x_{py} = \gamma pct_{py} + \alpha_p + \alpha_y + \varepsilon_{py}$ on subsamples of noncompliant and compliant facilities. Δx_{py} is a measure of the difference between demographic characteristic in counties up- and downstream from facility p in year y, and pct_{py} is a continuous variable that takes values from zero to one, and indicates the percent of downstream counties' populations living within a mile of a treated waterway in year y. The model includes facility and year fixed effects, α_p and α_y. Column 3 presents estimates of the associated triple difference, $\Delta x_{py} = \gamma^{DD} pct_{py} + \gamma^{DDD} pct_{py} * t_p + \alpha_y * t_p + \alpha_p + \alpha_y + \varepsilon_{py}$, where t_p is an indicator that equals one for noncompliant facilities. Each panel represents a different demographic variable. Means of each variable in 1970 from up- and downstream counties are reported at the bottom of each panel. Ninety-five percent confidence intervals in brackets. $^*p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 3
Effects on Health at Birth

	Full Sample (1)	Full Sample (2)	Full Sample (3)	Full Sample (4)	Noncompliant (5)	Compliant (6)	DDD (7)
Panel A: County Average Birth Weight							
Percent downstream	12.80***	6.718***	7.134***	8.999***	13.36***	5.153**	5.153**
	[6.709, 18.89]	[2.034, 11.40]	[2.444, 11.82]	[5.721, 12.28]	[8.012, 18.72]	[1.129, 9.177]	[1.130, 9.176]
Percent downstream × noncompliant							8.211**
							[1.519, 14.90]
Panel B: Probability Birth Weight < 2,500 Grams							
Percent downstream	−0.00288***	−0.000874*	−0.000963*	−0.00177***	−0.00216***	−0.00138**	−0.00138**
	[−0.00419, −0.00156]	[−0.00190, 0.000152]	[−0.00198, 0.0000584]	[−0.00256, −0.000985]	[−0.00334, −0.000979]	[−0.00244, −0.000325]	[−0.00244, −0.000325]
Percent downstream × noncompliant							−0.000780
							[−0.00236, 0.000803]
Unit and year fixed effects	X	X	X	X	X	X	X
Demographic controls		X	X	X	X	X	X
Up/downstream counties only			X	X	X	X	X
Collapsed to county level	X	X	X				
Collapsed to facility level				X	X	X	X
N	64,239	64,239	64,008	82,320	34,188	48,132	82,320

Source: National Center for Health Statistics (1968–1988a).

Notes: This table describes the effects of Clean Water Act grants on downstream infant health. In Panel A, the dependent variable is the average birth weight in a county–year, and in Panel B, it is the probability of being born weighing less than 2,500 grams. Columns 1–3 present estimates from Equation 2. Percent downstream is a continuous variable that takes values from zero to one and indicates the proportion of the population that lived within a mile of a treated waterway in each year. All estimates include unit and year fixed effects. Demographic controls include the percent of a county's births in a given birth order bin and county averages of mother's age and race and child gender. Columns 1 and 2 use data from every county in the United States, while Columns 3–7 restrict the sample to counties that are up- or downstream from a wastewater treatment facility. Columns 5 and 6 restrict the sample to noncompliant facilities (those that were required to make treatment technology upgrades) and compliant facilities (those that were not), respectively. In Columns 1–3, data are collapsed to the county level. In Columns 4–7, data are collapsed to the facility level. Columns 4–6 estimate Equation 3. Column 7 estimates the associated triple difference from Equation 4. All regressions are weighted by number of births. *$p < 0.10$, **$p < 0.05$, ***$p < 0.01$.

Equation 3, which compares birth weight in counties up- and downstream from the same facility. The point estimate is slightly larger in magnitude with a smaller confidence interval.[18,19]

The impact of the CWA on birth weight may not be uniform across the distribution of birth weight, so we also show results for low birth weight in Panel B of Table 3.[20] The point estimates are consistently negative, although not always significant, and range from −0.09 to −0.29 percentage points. About 7 percent of births in our sample are low birth weight, so this represents a change of 1–4 percent from the mean.

Finally, we estimate our triple difference specification on birth outcomes. Columns 5 and 6 of Table 3 present results from estimating Equation 3 on subsamples of noncompliant and compliant facilities, respectively. Consistent with our water pollution results in Table 1, we see a relatively large and statistically significant improvement in birth weight downstream from noncompliant facilities. The effect in areas downstream from compliant facilities is also positive, but smaller; improvements in birth weight in areas downstream from compliant facilities may be driven by demographic or economic changes that coincide with grant timing. Since, as shown in Table 2, demographic changes were similar in areas downstream from noncompliant and complaint facilities, the difference between the effects downstream from noncompliant and compliant facilities likely comes from the differences in surface water quality shown in Table 1, rather than shifting demographics.

Figures 2B and 2C present the event studies that correspond to our triple difference specification for birth weight and low birth weight, respectively.[21] There is no evidence of pre-treatment trends in infant health outcomes. For birth weight, there is a statistically significant increase in downstream (relative to upstream) counties after a noncompliant facility receives a grant (relative to other facilities).[22] For low birth weight, the point estimates are similar in shape but are less precise.

18. Online Appendix Figure A5 shows the associated event study figures for Column 4.

19. These results are identified off of comparisons of newly treated facilities relative to never-treated facilities, newly treated facilities relative to facilities that have not yet been treated, and newly treated facilities relative to already-treated facilities (Goodman-Bacon 2021). The third type of comparison can be wrong signed. We show in Online Appendix Sections B.2 and B.3 that our results are robust to using a stacked difference-in-difference design and Callaway and Sant'Anna (2020, 2021), which only rely on the first two types of comparisons. Online Appendix B.2 also provides estimates that use only the never-treated or not-yet-treated counterfactual comparisons, which highlights the different comparisons made in our main estimates.

20. Online Appendix Table A3 further explores the effect across the birth weight distribution. Replicating the specification from Column 2, we find reductions in low birth weight are driven by decreases in both extremely low birth weight (ELBW, defined as below 1,000 grams) and very low birth weight (VLBW, defined as below 1,500 grams). We also observe an increase in births greater than 2,500 grams, which suggests that there is an upward shift in the birth weight.

21. Online Appendix Figure A6 shows the birth weight event study figures for compliant and noncompliant facilities separately. Across both types of facilities, pre-treatment trends are very similar, and the effect of treatment is much larger for the noncompliant facilities in the post-treatment period, as expected.

22. In all of our event studies, we report coefficients for four years before and eight years after grant receipt, so that we only report balanced coefficients in our infant health specifications. These specifications also includes bins for five or more years before the grant and nine or more years after the grant, but our results are not sensitive to this choice of binning. While unbalanced event study coefficients should be interpreted with caution, Online Appendix Figure B2 presents a version of Figure 2B with 16 years of post-treatment data. This figure suggests that the effect of CWA grants on birth weight flattens out by ten years after treatment, consistent with grant projects taking up to ten years to complete (USEPA 2002).

We summarize the effect of changes in surface water quality downstream from noncompliant facilities on infant health by estimating Equation 4 on the pooled sample, which leverages all of our variation in one regression. Since Equation 4 includes a full set of interactions, our estimate of γ^{DDD}, reported in Column 7 of Table 3, is equivalent to the difference in the estimates in Columns 5 and 6. As in our water pollution estimates, the improvements in birth outcomes downstream from noncompliant facilities are statistically larger than improvements downstream from compliant facilities.[23] We show that this heterogeneity in effects is not driven by differences in facility size, population served, or nontreatment technology upgrades in Online Appendix Table B4, which provides further evidence that improvements in downstream infant health are driven by upgrades to treatment technology. In Online Appendix Section A.3, we also explore heterogeneity of the main results by maternal race and grant timing, but we find no significant differences along these dimensions.

The results from this triple difference show that increasing the probability of exposure to treated surface water from zero to 100 percent is associated with an 8.21-gram increase in average birth weight in counties downstream from facilities that were required to make upgrades to their treatment technology. In terms of magnitude, the effect on birth weight is about half of the estimated effect of any exposure to Ramadan during pregnancy (Almond and Mazumder 2011) and about the same magnitude as the effect of stress in utero due to nearby landmine explosions on birth weight (Camacho 2008). Estimates of the effect on the probability of low birth weight shown in Panel B of Table 3 are not significant, but they do bound improvements above a 0.236 percentage point decrease, or about 3 percent from the mean of low birth weight. This is slightly smaller than the estimated effect of drinking water contamination in utero on low birth weight estimated in a modern context (Currie et al. 2013).

C. Potential Mechanisms

There are several potential pathways through which surface water pollution could affect health, including contamination of public drinking water sources, contamination of private groundwater sources, and exposure through recreation. First, improvements in surface water quality from the CWA may affect infant health through a reduction in pollution in the source water that public water systems draw from. Public water systems, including those that draw from a surface water source, such as a lake or river, often violate health-based water quality standards, and these violations impact infant and child health (Currie et al. 2013; Grossman and Slusky 2019; Marcus 2022). A report by the U.S. Geological Survey (USGS) found that more than one in five source-water samples from public

23. It is important to note that the estimation equations for pollution and infant health are not identical, due to data limitations for the geography available in the infant health data, which only records county. For completeness, we also provide estimates of our first-stage impact on water quality using Equation 4 in Online Appendix Table A4. We find these estimates are statistically indistinguishable and reassuringly similar to our main specification. Online Appendix Figure A7 also shows that the associated event study reveals very similar dynamics to the health event studies.

water systems contained one or more contaminants at concentrations dangerous to human health. In an analysis of matched water samples from 94 water sources and their associated public water systems, the same organic contaminants detected in source water consistently appeared at similar concentrations in drinking water after treatment (Toccalino and Hopple 2010). In 1970, more than 70 percent of community water system users received drinking water from a surface water source (Dieter 2018), so improvements in surface water quality from CWA grants may have reduced exposure to harmful pollutants by improving public drinking water quality. Second, reductions in surface water pollution could affect populations who get their drinking water from private wells. Surface water quality can impact nearby groundwater quality through seepage and runoff, so private groundwater well users may also benefit from reduced pollution from wastewater treatment facilities. Since private wells are unregulated and untreated, these households may be at greater risk for direct exposure through their drinking water. Third, individuals could be exposed to surface water through recreation. This channel could impact health directly through physical contact with or ingestion of contaminated water or indirectly through changes in activity and exercise, such as swimming or walking along a waterway.

Our measure of treatment, the percent of a county's population that lives within one mile of a treated waterway, is likely to capture all three of these potential mechanisms to a certain extent. Impacts through private well contamination and through recreational exposure are likely to be largest for households nearest to the treated waterway. While our main results focus on the population within one mile of a treated waterway, it is not clear how far people will travel for water recreation, or how far private wells may be impacted.[24] With this in mind, Online Appendix Table A5 shows that our results are robust to using other bandwidths around treated waterways. We find a larger point estimate for a narrower bandwidth and a smaller yet still significant estimate for a wider bandwidth.

If public water is a primary channel of exposure, distance is not the ideal measure. Instead, a better measure of treatment would be the percent of the population served by a public water system drawing water from a treated waterway. Unfortunately, data limitations prevent us from creating this measure. Data on public and private water supply are very limited, especially in this historic context. Spatial data on public water system's service supply areas are only available for more recent time periods in a handful of states, and modern supply areas may not accurately reflect service areas in the 1970s and 1980s. In addition, we have no information on the exact location from which public water systems draw their water supply. There are also no historic data on private well locations. While not ideal, we still take advantage of the available spatial data on modern public water system's service areas from eight states for which these data are available (see Online Appendix Section C.2 for details on these data). For these states, we calculate the percent of a county's population

24. While distance is a factor in recreational use, accurately measuring recreational benefits is especially difficult (Kuwayama, Olmstead, and Zheng 2018). Unfortunately, we lack direct measures of water-related recreation activities from this time period.

living within a public water system's service area.[25] Online Appendix Table A7 shows that this measure, which is based on public water supply areas, is highly correlated with our primary treatment measure, which is based on distance. This suggests that our main results likely capture the public water channel. We also note that, to the extent there is measurement error in our measure of exposure, our estimates will be attenuated towards zero and could be interpreted as an underestimate.

These three mechanisms are difficult to measure and disentangle with available data. While these data limitations prevent us from definitively determining the main pathway of exposure, we explore potential mechanisms in Online Appendix Section A.5. We take advantage of country-wide data on the percent of a county's population receiving public water from groundwater or surface water sources in 1985. We find that our results are driven by counties with water systems drawing from surface water sources, which provides suggestive evidence that public drinking water sourced from surface waters is one channel through which reductions in surface water pollution can improve infant health.

VI. Conclusion

The preceding evidence suggests that the Clean Water Act led to small but significant improvements in birth weight, with reductions in pollution associated with CWA grants leading to an 8-gram increase in average birth weight in counties downstream from facilities that were required to make treatment technology upgrades. These results are consistent with the significantly larger improvements in water quality we find downstream from these facilities.

We note that our study has a number of important limitations. Data availability in this historic period restricts our ability to determine precise residential locations and the location of all drinking water sources and supply areas, as described in more detail in Section IV.C. To the extent that these data limitations generate measurement error in our estimates, the 8-gram increase in average birth weight may be understated.

Nevertheless, our estimates provide an important first step in understanding the effects of the Clean Water Act on health. Incorporating the birth weight benefits from cleaner surface water, as well as other infant, adolescent, and adult health effects, is important for a full accounting of the benefits of the CWA.[26] More generally, this research documents the importance of policies targeting cleaner water through sewage treatment in protecting health and shows that the complementarity between clean drinking water and sewerage initiatives for improving health holds well into the 20th century.

25. If populations are receiving publicly provided drinking water sourced upstream from other counties, our measure of treatment may not accurately describe the treated population. However, community water systems generally serve areas smaller than counties (USEPA 1997).

26. Although our study faces measurement error challenges and we cannot provide a full accounting of all relevant health benefits, we conduct a rough back-of-the-envelope calculation in Online Appendix Section A.6 to quantify the benefits of the effects on birth weight.